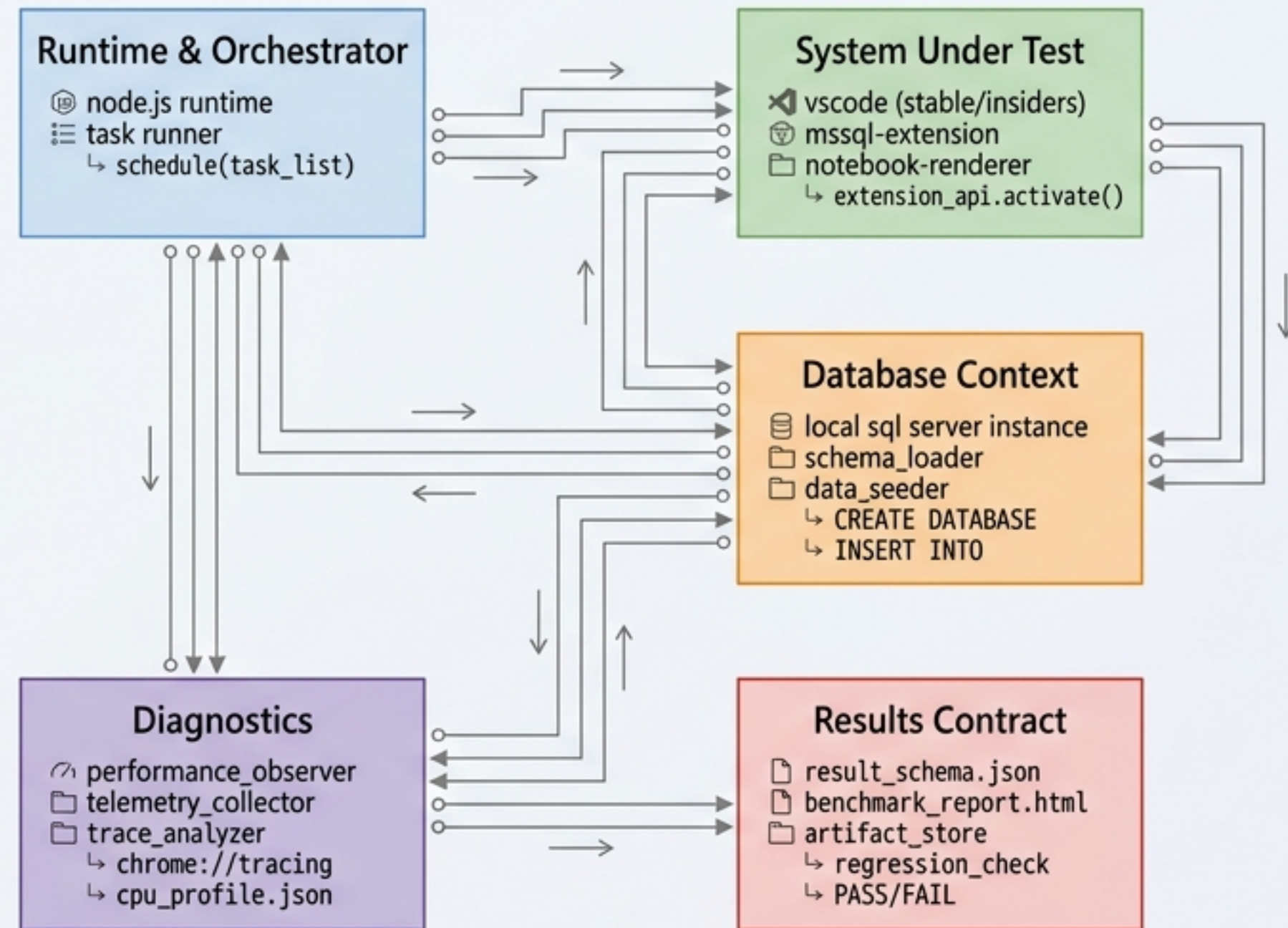


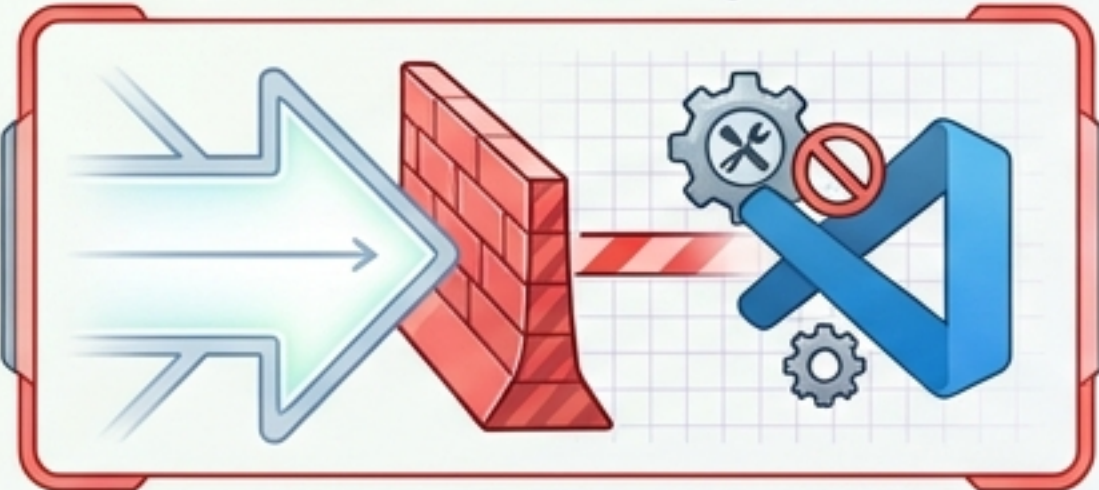
End-to-End Performance Harness for VS Code MSSQL

A local-first, deterministic system for measuring real user scenarios without forking the editor.



Core Principles and Architectural Guardrails

Guardrail 1: No Editor Forking



The harness relies entirely on **launch flags**, **CDP**, and **controlled extensions**. It measures the product shipping reality.

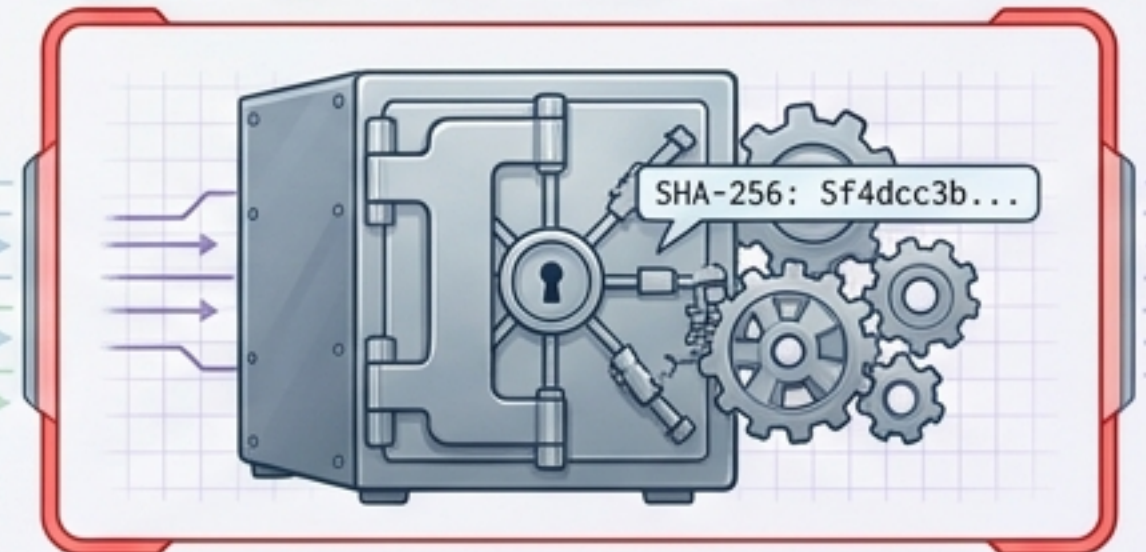
Guardrail 2: Zero Uncalibrated Observer Effect



Official metrics must never depend on ETW, CPU profiling, or heap snapshots. Missing collectors degrade diagnostic depth, not metric integrity.

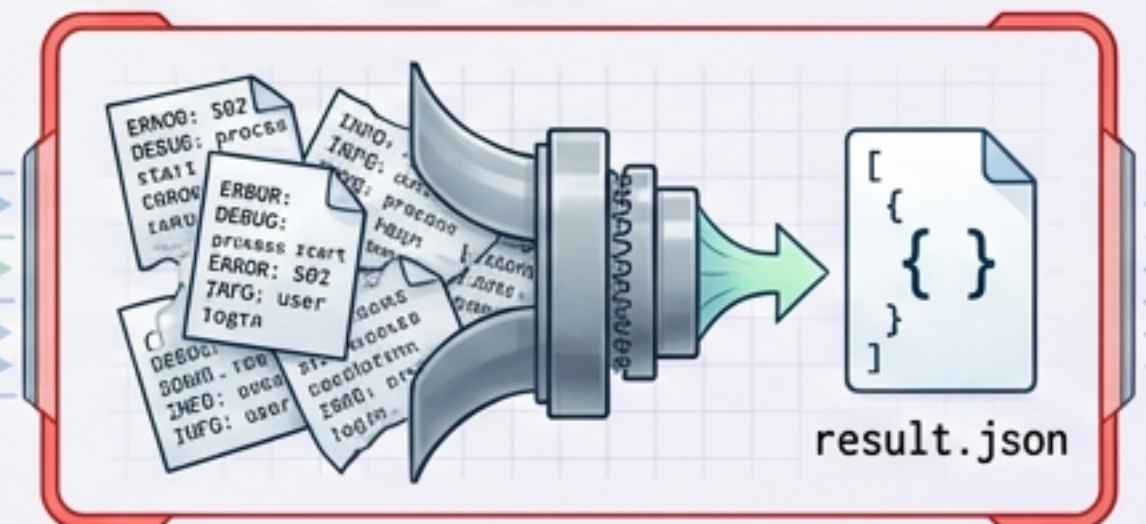
**Ideal
System
State**

Guardrail 3: Deterministic Environments



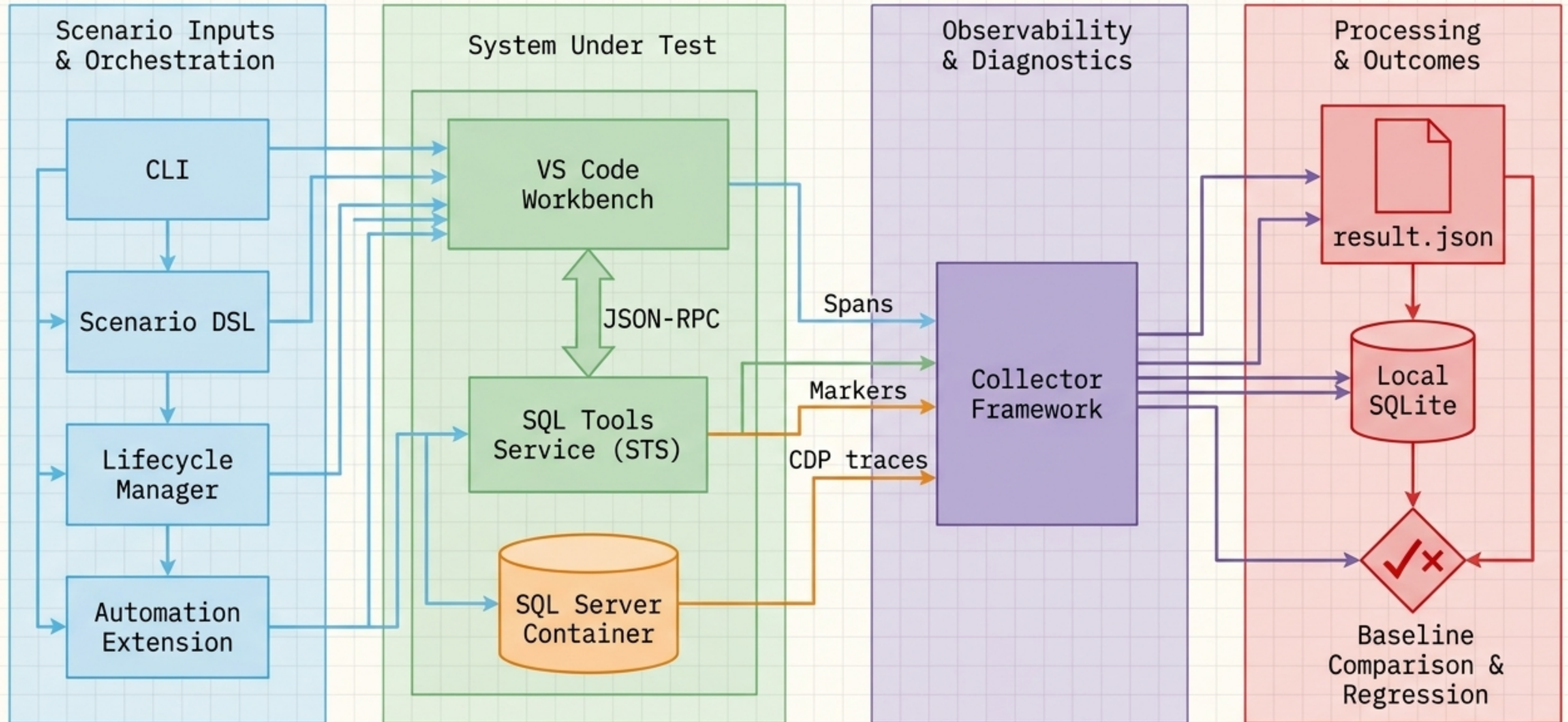
Every run must be perfectly **reproducible** via **config snapshots** and strict **hardware/environment fingerprinting**.

Guardrail 4: Canonical Data

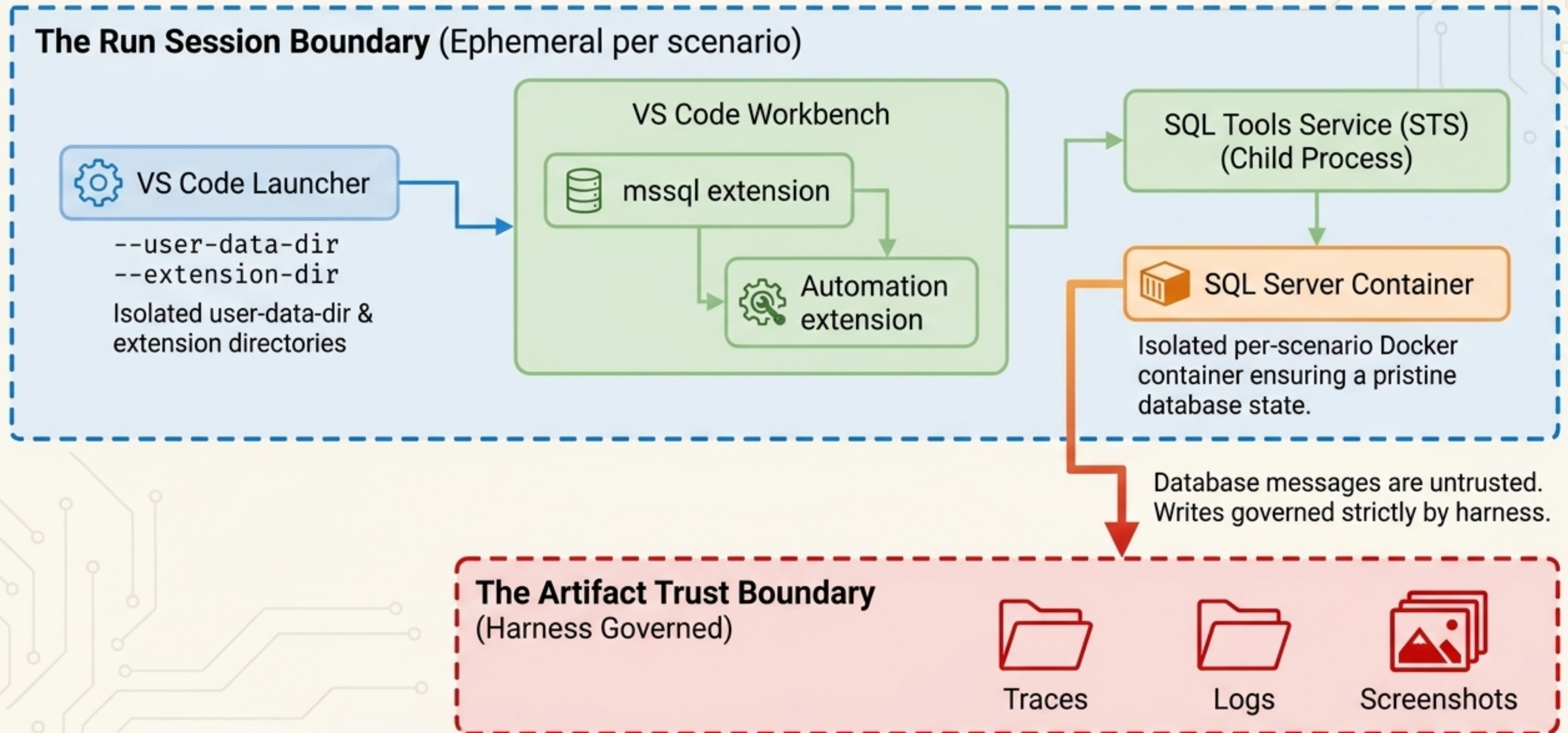


A stable per-rep contract keeps dashboards and regression logic decoupled from volatile internal implementations.

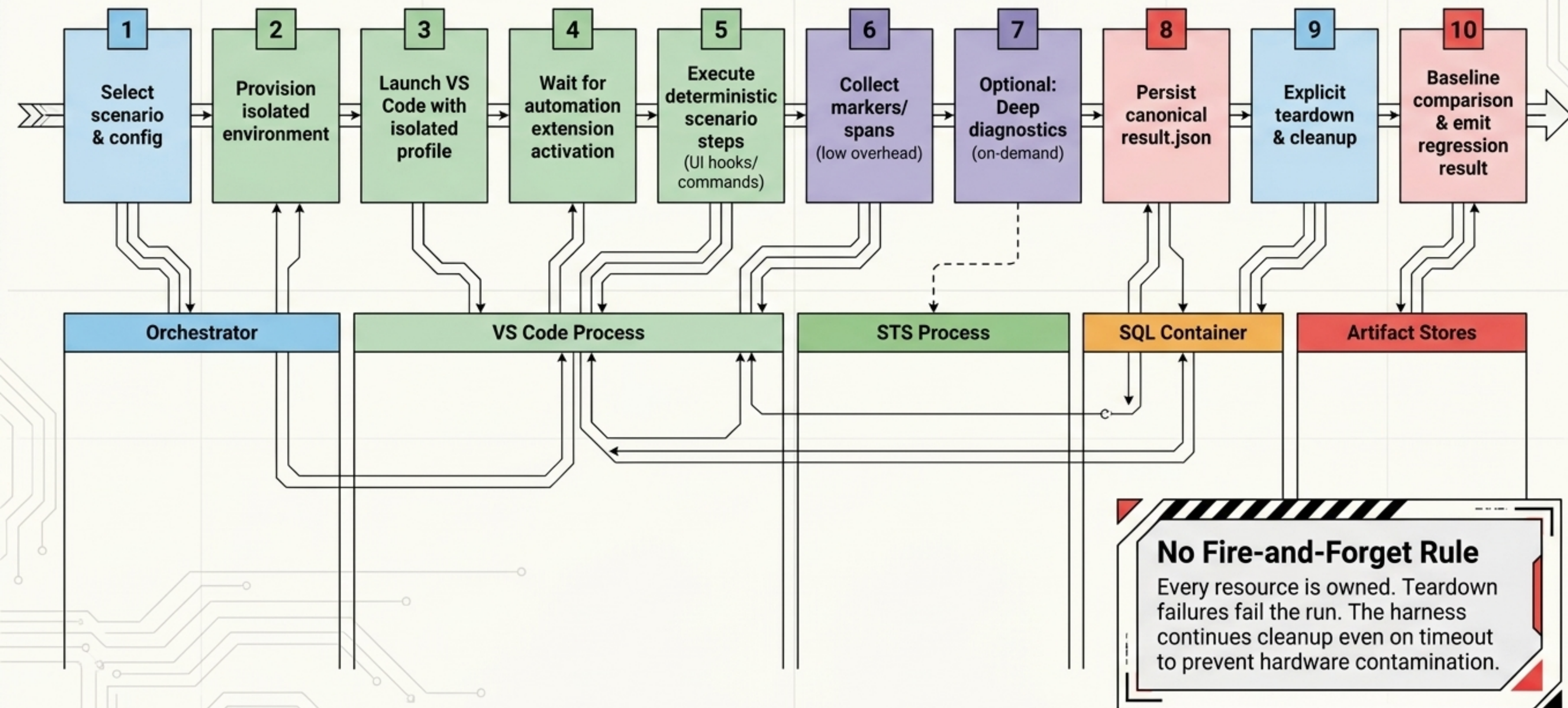
The System in Motion: From Configuration to Outcomes



Trust Boundaries and Process Isolation Strategy

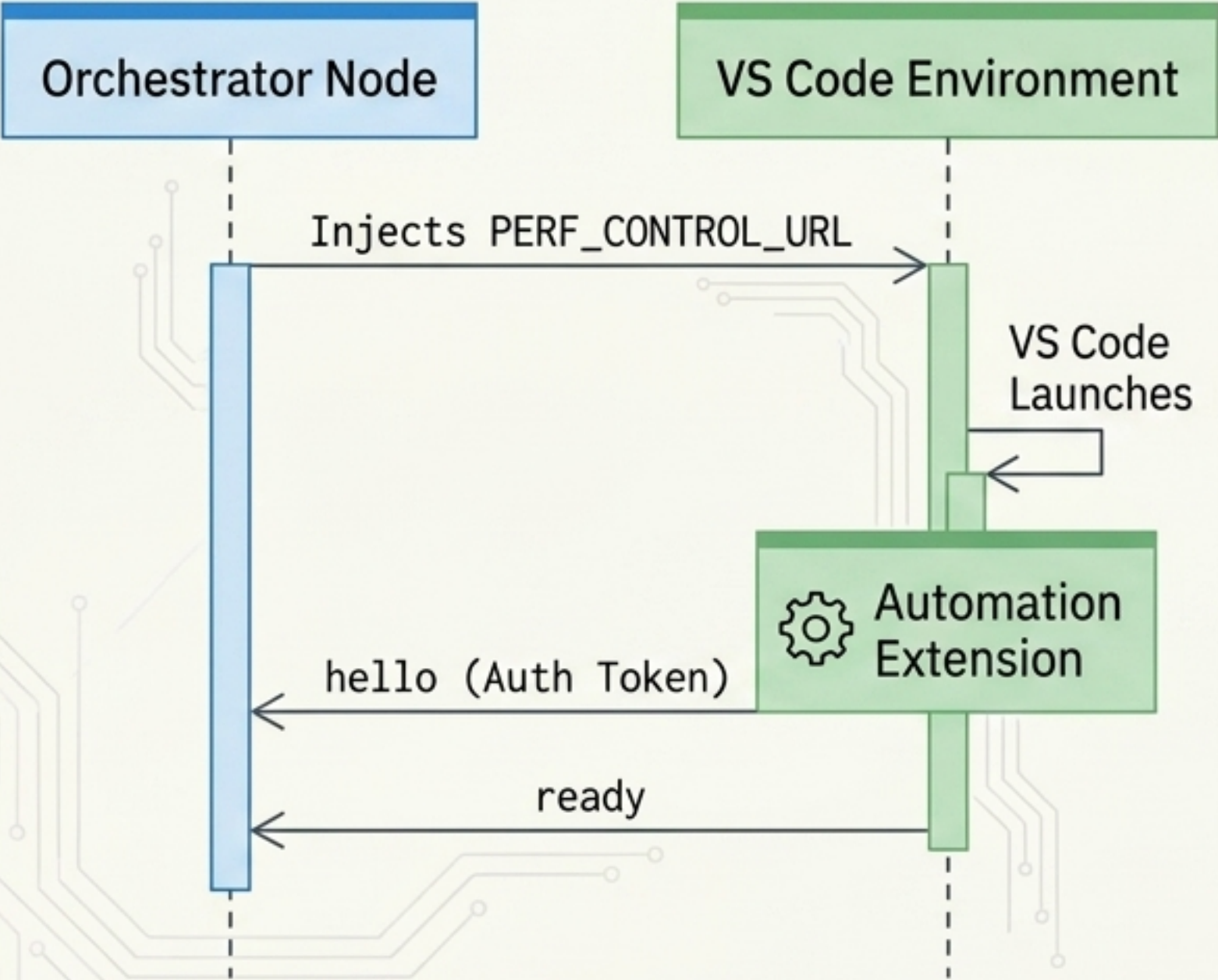


The 10-Step Controlled Scenario Lifecycle



The Automation Handshake and Scenario Profiles

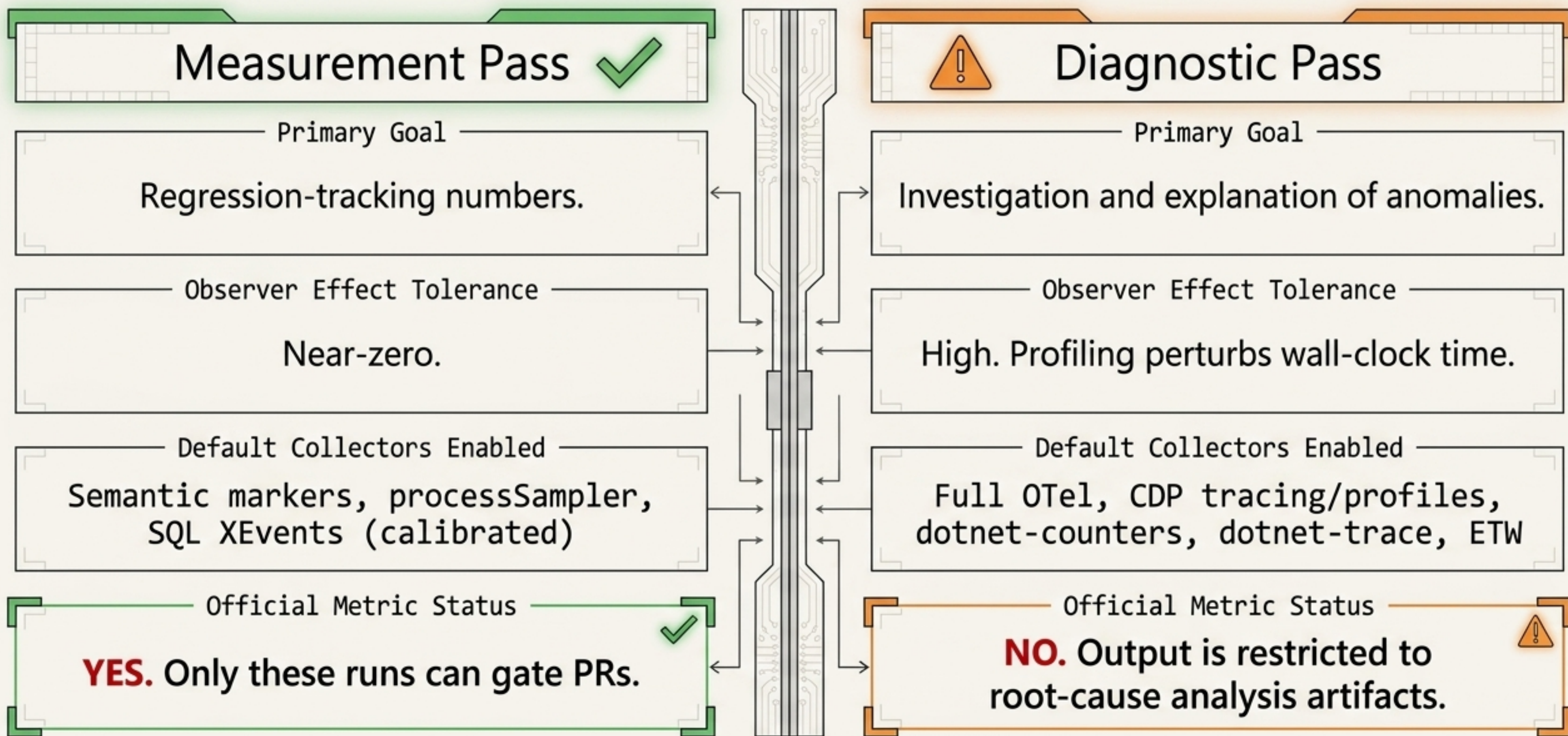
The Control Handshake



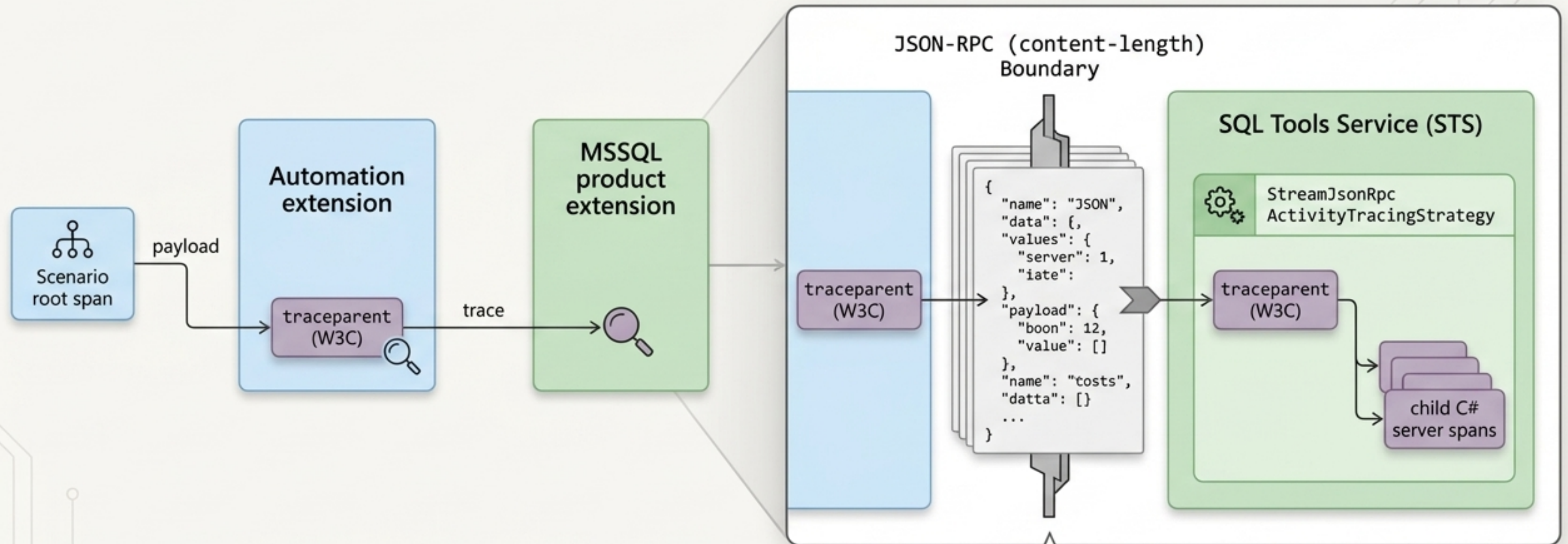
Scenario Mode Matrix

Profile Mode & Description	Target Scenario & Use Case
Fresh Mode New user/extension dirs per rep.	First-launch/install tests.
Warmed Mode Created once during setup, then copied/reused.	Normal activation, query execution, Object Explorer rendering.
Reuse Mode Same state across reps.	Exploratory diagnostics only (not official).

The Measurement Dichotomy: Regression vs. Root-Cause

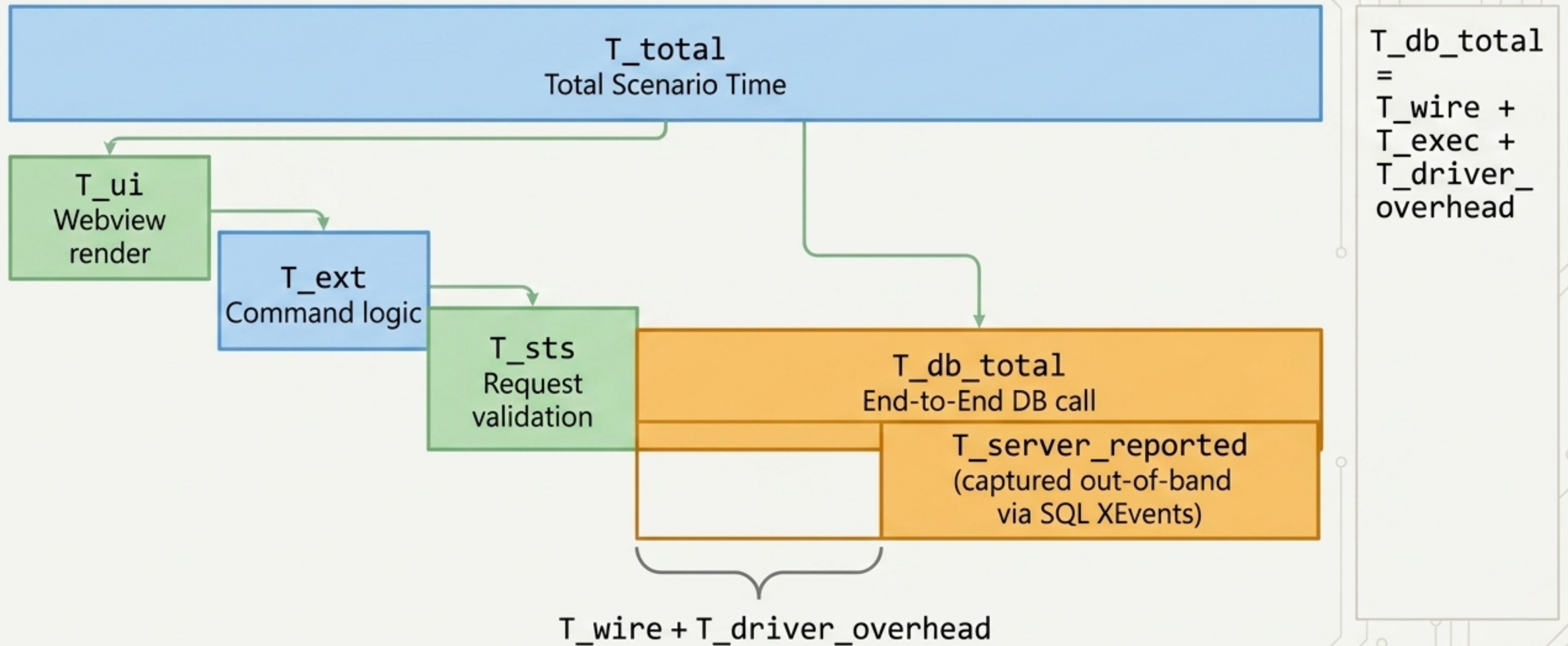


The Architectural Seam: Trace Propagation Across Processes

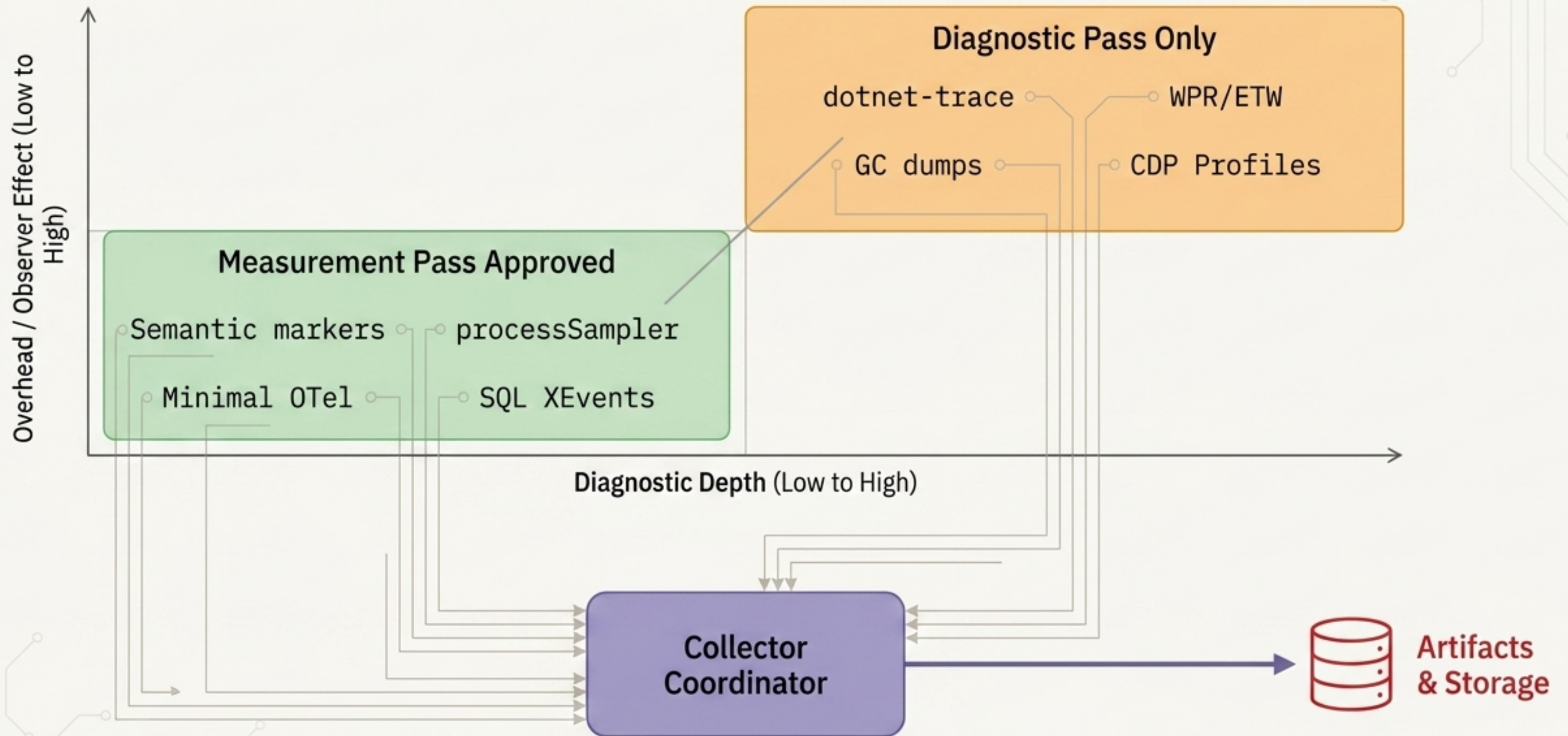


The highest-leverage observability seam: it is the only reliable boundary where extension and service clocks can be correlated with a known transport and payload size.

Timing Decomposition: Isolating SQL Server Execution

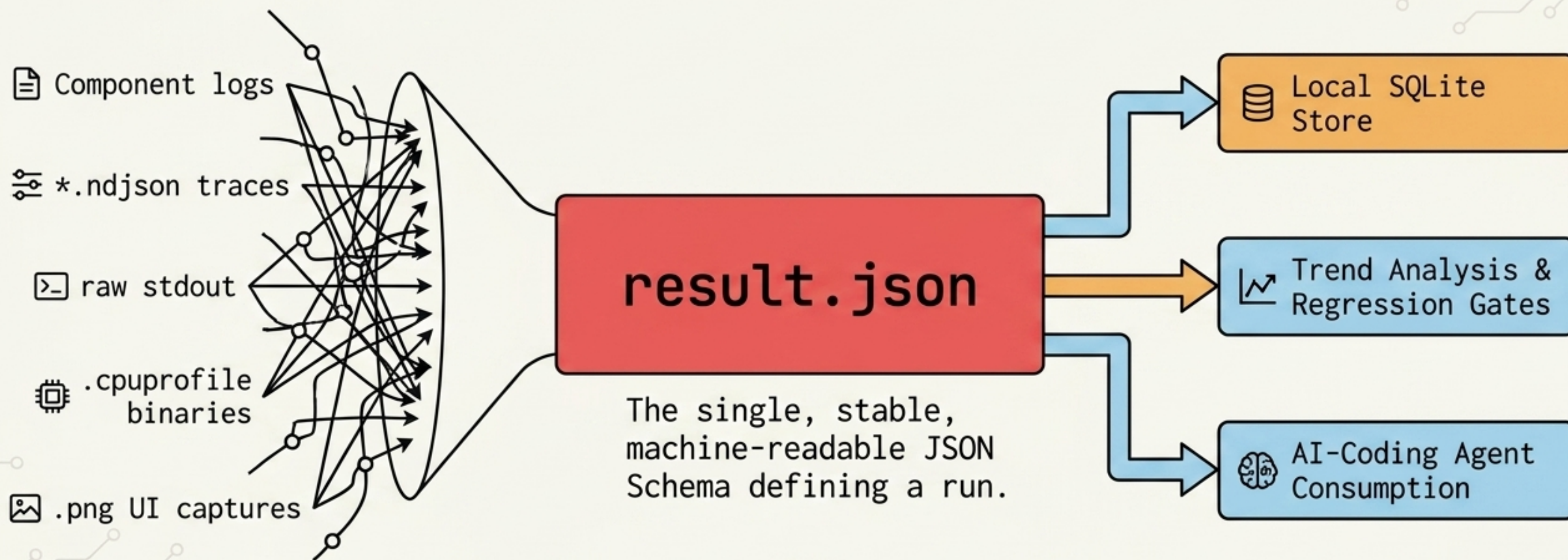


The Collector Catalog and Routing Framework



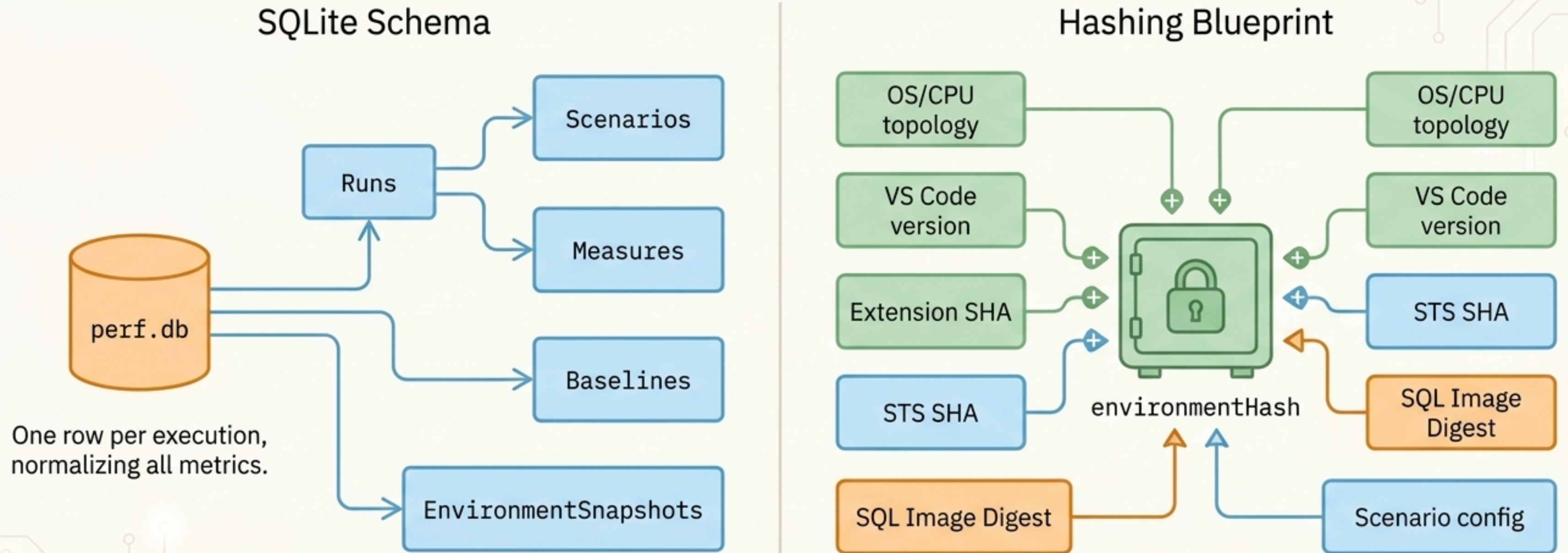
Collectors MUST NOT stall the run or alter timing behavior.
Routing is non-blocking and back-pressure-aware.

The Canonical Contract: `result.json`



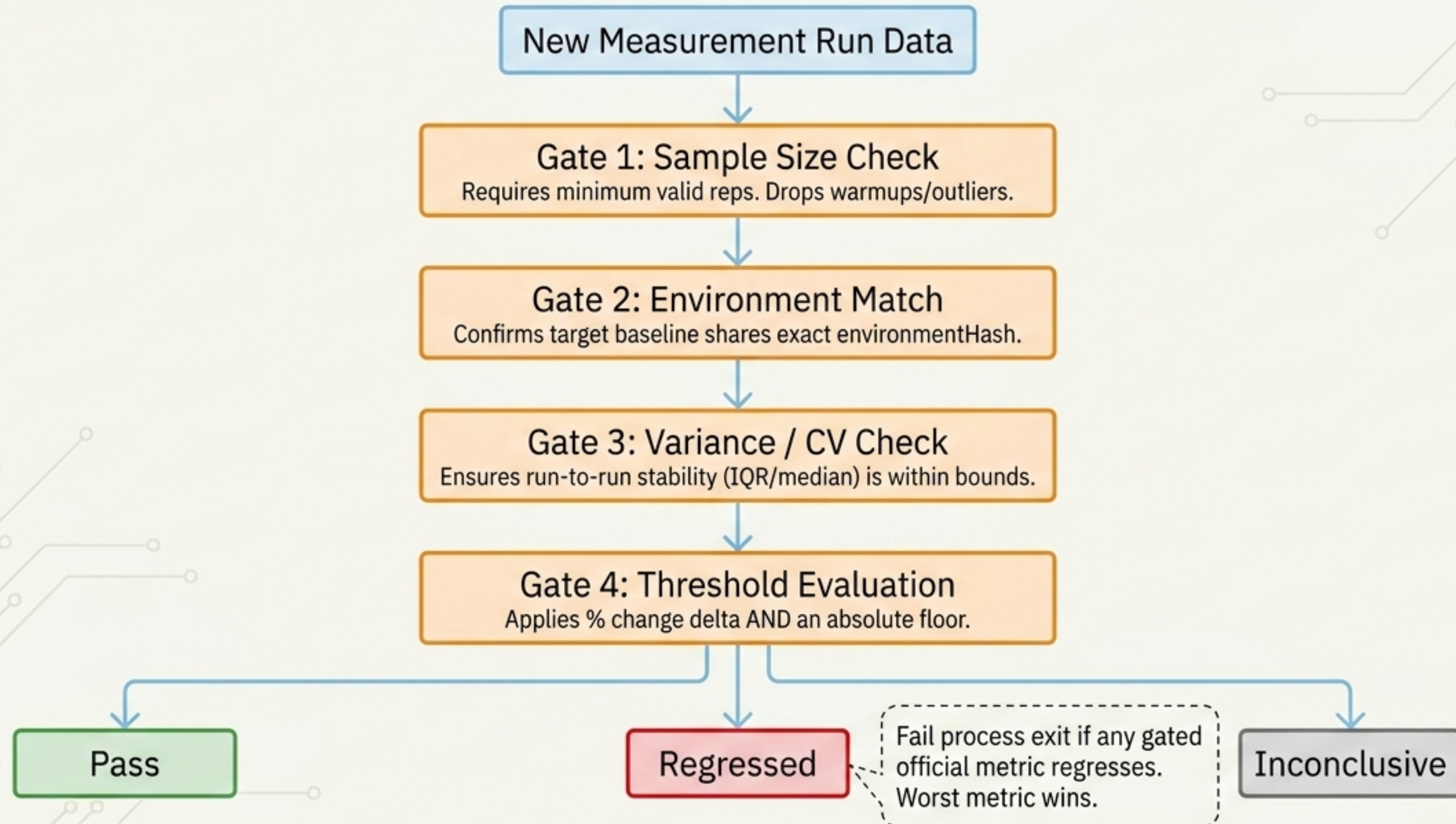
As long as the contract is stable and versioned, internal implementations can evolve without breaking downstream regression tools.

Local-First Storage and Environment Fingerprinting

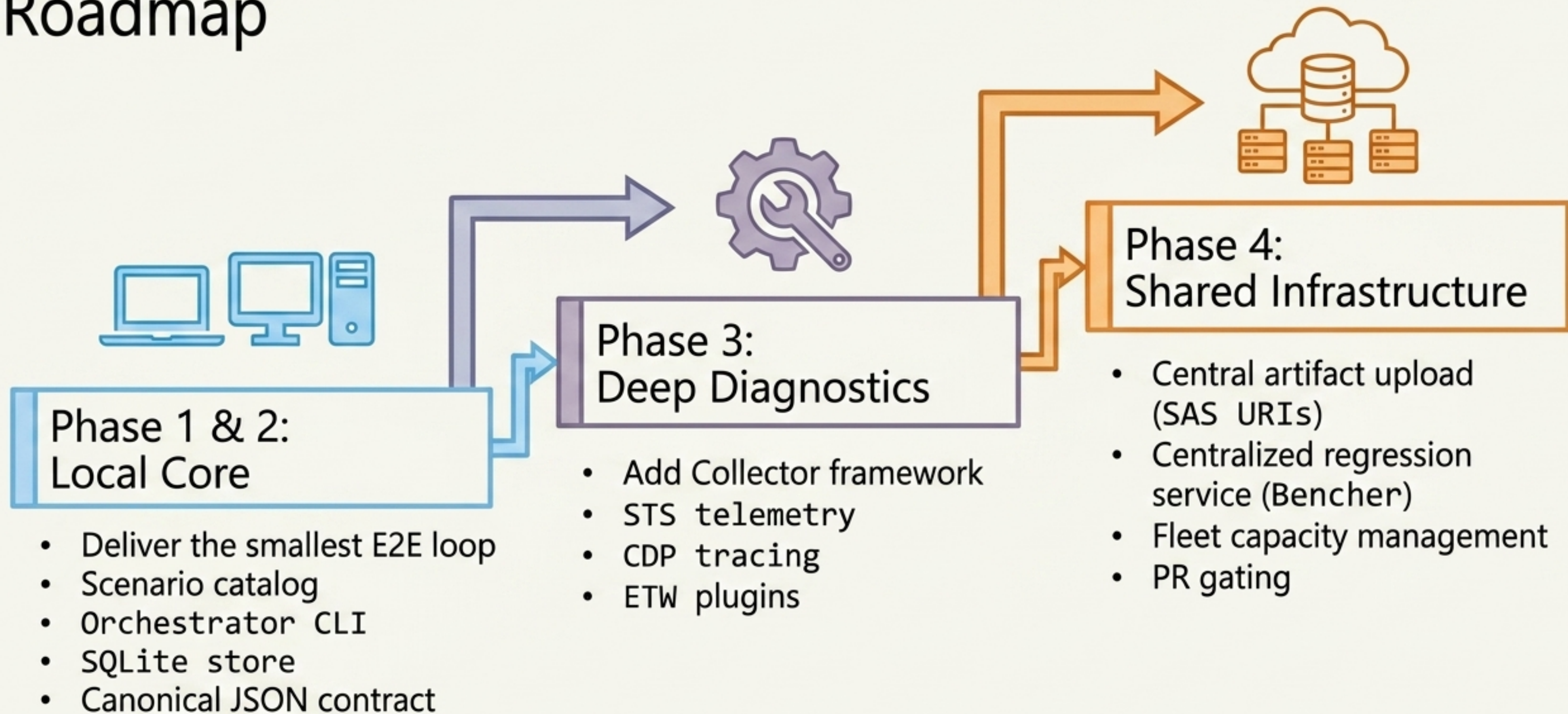


The Golden Rule: Never compare official metrics across different environment hashes unless explicitly configured.

The Statistical Gate: Automated Regression Analysis

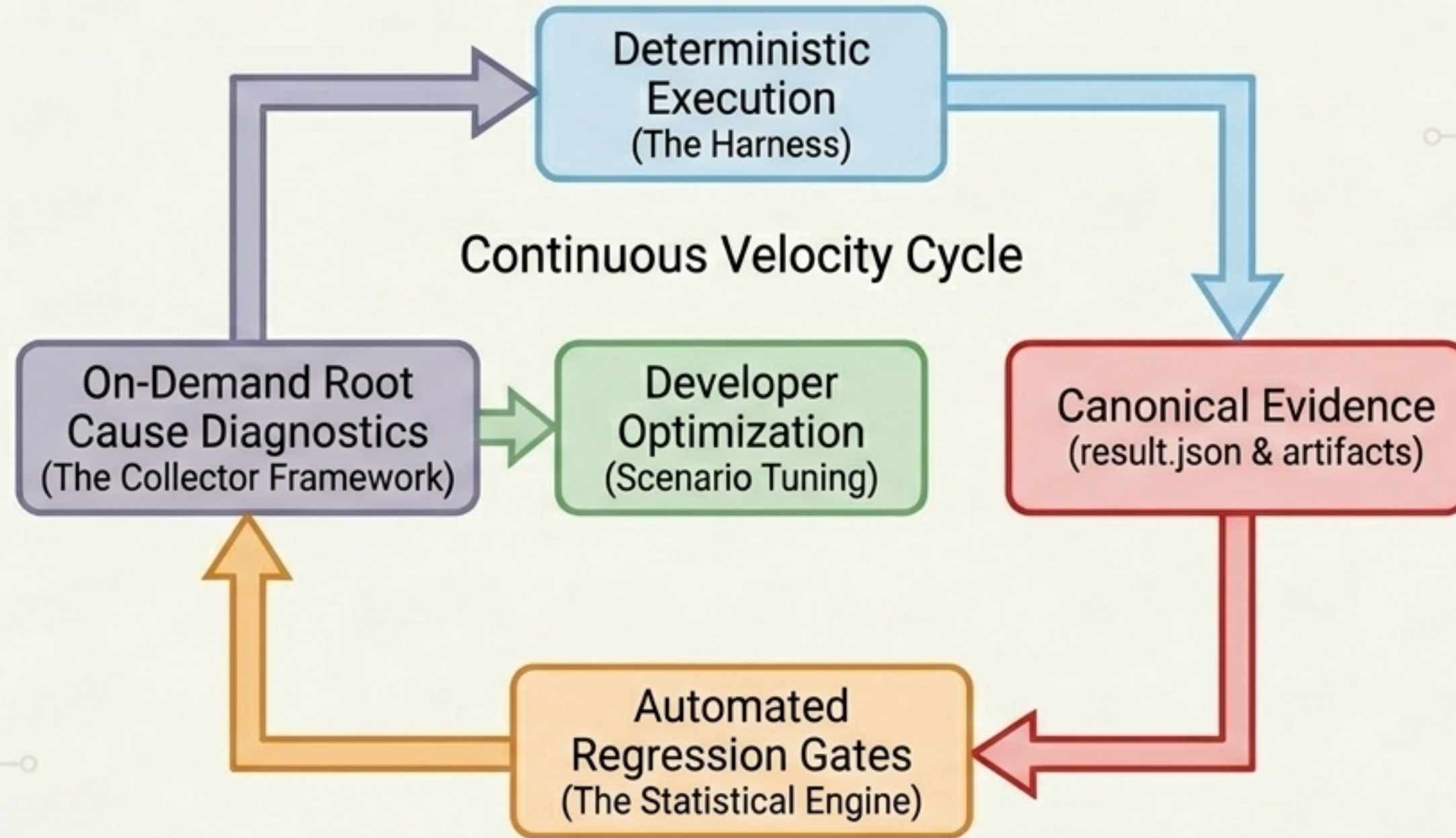


Deployment Model and Implementation Roadmap



Hardware ownership matters.

Synthesis: The Ultimate Feedback Loop



By separating cheap measurement from heavy diagnostics, anchoring traceability via JSON-RPC, and strictly isolating environments, the VS Code MSSQL performance harness transforms performance testing from anecdotal observation into a repeatable, scientific discipline.